# PrivLens: Privacy Evidence & Metadata Inventory

Empirical PII Discovery, RoPA Governance Audit & Benchmark Report | Generated: October 27, 2026 • 22:44 UTC

**LEGAL BOUNDARY & AUDIT NOTICE:** This technical report was compiled through automated syntactic discovery and column profiling. It represents factual technical evidence intended for privacy steward and DPO review. It **does NOT** constitute an automated lawful-basis decision under GDPR/CCPA, nor does it deduce universal cross-border transfer topologies.

## 1. Executive Technical Summary

| Metric Name | Count / Value | Context & Classification Scope |
|---|---|---|
| **Total Fields Scanned** | **44** | Structured CSV spreadsheets and SQLite database tables |
| **Candidate PII Discovered** | **31** | Personal data fields across 5 predefined data classes |
| Non-PII Columns | 13 | Technical keys, surrogate IDs, timestamps, status flags |
| **Governance & Evidence Gaps** | **15** | Unregistered shadow data, missing purposes, or missing retention |

### Distribution Across 5 Predefined Data Classes

| Data Class Name | Candidate Fields | Scope & Subtypes Observed |
|---|---|---|
| **Personal identity (PID)** | 6 | Name, date of birth, gender, age |
| **Contact information (CNT)** | 4 | Email address, phone number |
| **Government identifiers (GID)** | 7 | Aadhaar, PAN, passport, voter ID, driving licence |
| **Financial information (FIN)** | 8 | Bank account number, IFSC, card number, UPI ID |
| **Location / address (LOC)** | 6 | Postal address, PIN code, geo-coordinates |

## 2. Structured Technical Evidence Inventory

| Target Asset / Column | Discovered Class | Conf. | Declared RoPA Purpose | Retention | Status |
|---|---|---|---|---|---|
| customer_signups.csv **customer_name** | Personal identity | **100%** | Customer Order Fulfillment & Shipping | 36m (Account Deletion) | **ACTIVE** |
| customer_signups.csv **email_address** | Contact information | **100%** | Customer Order Fulfillment & Shipping | 36m (Account Deletion) | **ACTIVE** |
| customer_signups.csv **phone_number** | Contact information | **100%** | Customer Order Fulfillment & Shipping | 24m (Order Completion) | **ACTIVE** |
| customer_signups.csv **date_of_birth** | Personal identity | **100%** | *Missing Purpose* | *Missing* | **UNREGISTERED** |
| customer_signups.csv **gender** | Personal identity | **100%** | *Missing Purpose* | 12m (Account Review) | **UNLINKED_PURPOSE** |
| customer_signups.csv **residential_address** | Location / address | **100%** | Customer Order Fulfillment & Shipping | *Missing* | **INCOMPLETE** |
| customer_signups.csv **pin_code** | Location / address | **100%** | Customer Order Fulfillment & Shipping | 36m (Account Deletion) | **ACTIVE** |
| digital_payments_log.csv **bank_account_number** | Financial information | **100%** | Payment Processing & Financial Ledger | 84m (Financial Year End) | **ACTIVE** |
| digital_payments_log.csv **ifsc_code** | Financial information | **100%** | Payment Processing & Financial Ledger | 84m (Financial Year End) | **ACTIVE** |
| digital_payments_log.csv **card_number** | Financial information | **100%** | *Missing Purpose* | 24m (Card Expiration) | **UNLINKED_PURPOSE** |
| digital_payments_log.csv **upi_id** | Financial information | **100%** | *Missing Purpose* | *Missing* | **UNREGISTERED** |
| kyc_verification_records.csv **aadhaar_number** | Government identifiers | **100%** | Statutory KYC & Customer Verification | 60m (Statutory Audit Window) | **ACTIVE** |
| kyc_verification_records.csv **pan_number** | Government identifiers | **100%** | Statutory KYC & Customer Verification | 60m (Statutory Audit Window) | **ACTIVE** |
| kyc_verification_records.csv **passport_number** | Government identifiers | **100%** | Statutory KYC & Customer Verification | *Missing* | **INCOMPLETE** |
| kyc_verification_records.csv **voter_id** | Government identifiers | **100%** | *Missing Purpose* | *Missing* | **UNREGISTERED** |
| kyc_verification_records.csv **driving_licence** | Government identifiers | **100%** | *Missing Purpose* | *Missing* | **UNREGISTERED** |
| enterprise_data.db (employees) **employee_name** | Personal identity | **100%** | Employee Payroll & Statutory Benefits | 84m (Termination of Employment) | **ACTIVE** |
| enterprise_data.db (employees) **age** | Personal identity | **100%** | Employee Payroll & Statutory Benefits | 84m (Termination of Employment) | **ACTIVE** |
| enterprise_data.db (employees) **pan_card** | Government identifiers | **100%** | Employee Payroll & Statutory Benefits | 84m (Statutory Tax Filing) | **ACTIVE** |
| enterprise_data.db (employees) **salary_account_no** | Financial information | **100%** | Employee Payroll & Statutory Benefits | 84m (Termination of Employment) | **ACTIVE** |
| enterprise_data.db (employees) **bank_ifsc** | Financial information | **100%** | Employee Payroll & Statutory Benefits | 84m (Termination of Employment) | **ACTIVE** |
| enterprise_data.db (delivery_dispatch) **delivery_address** | Location / address | **100%** | Customer Order Fulfillment & Shipping | 12m (Delivery Confirmation) | **ACTIVE** |
| enterprise_data.db (delivery_dispatch) **postal_pin** | Location / address | **100%** | Customer Order Fulfillment & Shipping | 12m (Delivery Confirmation) | **ACTIVE** |

| Target Asset / Column | Discovered Class | Conf. | Declared RoPA Purpose | Retention | Status |
|---|---|---|---|---|---|
| enterprise_data.db (delivery_dispatch) **geo_coords** | Location / address | **100%** | Customer Order Fulfillment & Shipping | *Missing* | **INCOMPLETE** |
| enterprise_data.db (delivery_dispatch) **order_upi** | Financial information | **100%** | Payment Processing & Financial Ledger | 24m (Dispute Window) | **ACTIVE** |
| enterprise_data.db (legacy_unindexed_vault) **aadhaar_card** | Government identifiers | **100%** | *Missing Purpose* | *Missing* | **UNREGISTERED** |
| enterprise_data.db (legacy_unindexed_vault) **caller_phone** | Contact information | **100%** | *Missing Purpose* | *Missing* | **UNREGISTERED** |
| enterprise_data.db (legacy_unindexed_vault) **contact_email** | Contact information | **100%** | *Missing Purpose* | *Missing* | **UNREGISTERED** |
| enterprise_data.db (legacy_unindexed_vault) **card_number** | Financial information | **100%** | *Missing Purpose* | *Missing* | **UNREGISTERED** |
| enterprise_data.db (legacy_unindexed_vault) **birth_date** | Personal identity | **100%** | *Missing Purpose* | *Missing* | **UNREGISTERED** |
| enterprise_data.db (legacy_unindexed_vault) **geo_coordinates** | Location / address | **100%** | *Missing Purpose* | *Missing* | **UNREGISTERED** |

## 3. Evidence Gaps & Missing Metadata Action Checklist

| Sev. | Target Field Identifier | Gap Type | Evidence Findings & Recommended Human Action |
|---|---|---|---|
| CRITICAL | `customer_signups.csv::date_of_birth` | UNREGISTERED_SHADOW_DATA | **Unregistered Candidate PII in customer_signups.csv**: Field 'date_of_birth' was identified as candidate personal data (Personal identity, confidence: 100%) across 150 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning. |
| CRITICAL | `digital_payments_log.csv::upi_id` | UNREGISTERED_SHADOW_DATA | **Unregistered Candidate PII in digital_payments_log.csv**: Field 'upi_id' was identified as candidate personal data (Financial information, confidence: 100%) across 100 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning. |
| CRITICAL | `kyc_verification_records.csv::voter_id` | UNREGISTERED_SHADOW_DATA | **Unregistered Candidate PII in kyc_verification_records.csv**: Field 'voter_id' was identified as candidate personal data (Government identifiers, confidence: 100%) across 120 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning. |
| CRITICAL | `kyc_verification_records.csv::driving_licence` | UNREGISTERED_SHADOW_DATA | **Unregistered Candidate PII in kyc_verification_records.csv**: Field 'driving_licence' was identified as candidate personal data (Government identifiers, confidence: 100%) across 120 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning. |
| CRITICAL | `enterprise_data.db:legacy_unindexed_vault::aadhaar_card` | UNREGISTERED_SHADOW_DATA | **Unregistered Candidate PII in enterprise_data.db:legacy_unindexed_vault**: Field 'aadhaar_card' was identified as candidate personal data (Government identifiers, confidence: 100%) across 70 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning. |
| CRITICAL | `enterprise_data.db:legacy_unindexed_vault::caller_phone` | UNREGISTERED_SHADOW_DATA | **Unregistered Candidate PII in enterprise_data.db:legacy_unindexed_vault**: Field 'caller_phone' was identified as candidate personal data (Contact information, confidence: 100%) across 70 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning. |
| CRITICAL | `enterprise_data.db:legacy_unindexed_vault::contact_email` | UNREGISTERED_SHADOW_DATA | **Unregistered Candidate PII in enterprise_data.db:legacy_unindexed_vault**: Field 'contact_email' was identified as candidate personal data (Contact information, confidence: 100%) across 70 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning. |

| Sev. | Target Field Identifier | Gap Type | Evidence Findings & Recommended Human Action |
|---|---|---|---|
| CRITICAL | `enterprise_data.db:legacy_unindexed_vault::card_number` | UNREGISTERED_SHADOW_DATA | **Unregistered Candidate PII in enterprise_data.db:legacy_unindexed_vault**: Field 'card_number' was identified as candidate personal data (Financial information, confidence: 100%) across 70 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning. |
| CRITICAL | `enterprise_data.db:legacy_unindexed_vault::birth_date` | UNREGISTERED_SHADOW_DATA | **Unregistered Candidate PII in enterprise_data.db:legacy_unindexed_vault**: Field 'birth_date' was identified as candidate personal data (Personal identity, confidence: 100%) across 70 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning. |
| CRITICAL | `enterprise_data.db:legacy_unindexed_vault::geo_coordinates` | UNREGISTERED_SHADOW_DATA | **Unregistered Candidate PII in enterprise_data.db:legacy_unindexed_vault**: Field 'geo_coordinates' was identified as candidate personal data (Location / address, confidence: 100%) across 70 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning. |
| HIGH | `customer_signups.csv::gender` | MISSING_PURPOSE | **Missing Declared Purpose for gender**: Field 'gender' is registered under customer_signups.csv but has no declared business processing purpose. ■ **Action:** Privacy Steward Action: Assign and validate a declared business purpose prior to lawful processing verification. |
| HIGH | `customer_signups.csv::residential_address` | MISSING_RETENTION | **Missing Retention Schedule for residential_address**: Field 'residential_address' contains candidate personal data without a defined retention limit or deletion trigger. ■ **Action:** Data Governance Action: Formulate data retention schedule and disposal trigger in accordance with corporate data retention policy. |
| HIGH | `digital_payments_log.csv::card_number` | MISSING_PURPOSE | **Missing Declared Purpose for card_number**: Field 'card_number' is registered under digital_payments_log.csv but has no declared business processing purpose. ■ **Action:** Privacy Steward Action: Assign and validate a declared business purpose prior to lawful processing verification. |
| HIGH | `kyc_verification_records.csv::passport_number` | MISSING_RETENTION | **Missing Retention Schedule for passport_number**: Field 'passport_number' contains candidate personal data without a defined retention limit or deletion trigger. ■ **Action:** Data Governance Action: Formulate data retention schedule and disposal trigger in accordance with corporate data retention policy. |
| HIGH | `enterprise_data.db:delivery_dispatch::geo_coords` | MISSING_RETENTION | **Missing Retention Schedule for geo_coords**: Field 'geo_coords' contains candidate personal data without a defined retention limit or deletion trigger. ■ **Action:** Data Governance Action: Formulate data retention schedule and disposal trigger in accordance with corporate data retention policy. |

## 4. Benchmark Precision & Recall Performance

Evaluated across **44 fields** against ground truth. Macro Precision: **100.0%** | Macro Recall: **100.0%** | Macro F1-Score: **100.0%**

| Code | Predefined Data Class | Support | TP | FP | FN | Precision | Recall | F1-Score |
|---|---|---|---|---|---|---|---|---|
| **PID** | Personal identity | 6 | 6 | 0 | 0 | 100.0% | 100.0% | **100.0%** |
| **CNT** | Contact information | 4 | 4 | 0 | 0 | 100.0% | 100.0% | **100.0%** |
| **GID** | Government identifiers | 7 | 7 | 0 | 0 | 100.0% | 100.0% | **100.0%** |
| **FIN** | Financial information | 8 | 8 | 0 | 0 | 100.0% | 100.0% | **100.0%** |
| **LOC** | Location / address | 6 | 6 | 0 | 0 | 100.0% | 100.0% | **100.0%** |